

Problem A. AND Permutation

Input file: *standard input*
Output file: *standard output*
Time limit: 5 seconds
Memory limit: 1024 mebibytes

You are given a sequence of n distinct nonnegative integers $a_1, a_2, \dots, a_n$.

For the given sequence, it is guaranteed that for all nonnegative numbers x , if there is some i such that $a_i \& x = x$, then there is a j such that $a_j = x$. Here, $\&$ refers to the bitwise AND operator.

Find a permutation $b_1, b_2, \dots, b_n$ of $a_1, a_2, \dots, a_n$ such that $b_i \& a_i = 0$ for all i . If there are multiple solutions, find any such permutation. It is guaranteed that a solution always exists.

Input

The first line of input contains an integer n ($1 \leq n < 2^{18}$), which is the number of integers in the permutation.

Each of the next n lines contains an integer a_i ($0 \leq a_i < 2^{60}$), which is the input sequence, in order of i . All of the a_i 's are guaranteed to be distinct. For all nonnegative numbers x , if there is some i such that $a_i \& x = x$, then there is a j such that $a_j = x$.

Output

Output n lines, each containing a single integer, which are the b_i 's, in order of i .

Example

standard input	standard output
6	4
0	6
1	0
4	2
5	5
2	1
6	